

Development Summary — The Cottages at Arcade Springs

Rezoning application, R-1 → R-2 (Medium-Density Residential), City of Lilburn, Georgia **Application item:** Development Summary / site data report (companion to the Letter of Intent and Master Concept Plan) **Status:** DRAFT — owner-prepared, 2026-09-03. Boundary, topography, stream location, sewer inverts, stormwater volumes and traffic figures are preliminary and are to be superseded by sealed work of a Georgia RLS, PE, architect and landscape architect. **Basis documents:** FACTS.md (site geometry and code standards; §2 corrected 2026-09-03); data/layout.json (Master Concept Plan generator, **41 lots**, regenerated 2026-09-03); data/plans.json (architectural envelopes, regenerated 2026-09-03); the three re-verification audits at audit-2026-09-03/ (site-geometry.md, external-facts.md, docs-consistency.md); docs/11-site-context-brief.md; Lilburn Zoning Ordinance, Ord. 2023-603; Lilburn 2026 Application Instructions — Rezoning/SUP/CIC; RZ-2025-01 packet (status/rz-2025-01/). Derived numbers are computed in tools/dev_summary_calcs.py → data/dev-summary-derived.json, each one recomputed from the raw geometry and then cross-checked against the generator's own summary block. Since the generator was corrected on 2026-09-03 the two agree on every tracked quantity within 2 %; the three residual differences (all under 0.1 %) are stated where they occur rather than hidden.

Lilburn does not require the §736 Mixed-Use development report for an R-2 application; this summary is offered voluntarily so that staff has the same content in the same order.

Local coordinates used below: **u** = feet along the strip's long axis from the Arcade Rd right-of-way corner (bearing N 28°43' W toward the NW end); **v** = feet across the strip, positive toward the NE line, negative toward the SW line (FACTS §1).

Every finding in this document is stated as **"appears consistent with"** the cited standard. Nothing here determines that the project meets the code; that determination is the reviewing official's to make.

What changed on 2026-09-03, and what it cost

The site-plan generator was corrected against audit-2026-09-03/site-geometry.md. **The lot yield fell from 43 to 41**, for two reasons, both of them the applicant's own screening finding rather than a staff comment:

- 1. **Every lot is now screened against all three digitised stream reaches, not only the on-site one.** The southwest slot at u 1,230–1,280 failed the screen and is no longer lotted; it is a stream-setback open-space tract of 5,000 SF. **Cost: one lot.**
- 2. **The detention basins were sized on an assumed 4.4 disturbed acres. The real disturbed area is 7.541 ac**, so the required screening volume rose from ≈ 44,000 cf to **75,410 cf** and both basins had to be enlarged — Pond 2's tract from 15,000 to 20,000 SF. **Cost: one lot.**

Everything derived from the count moves with it: density, parking, trips, sewer flow, the tax base, the phase split, the HOPA 80-percent computation and the HOA budget. Two things got **better**: with the turnarounds re-spaced, the longest block is now **594 ft against the 600-ft standard of Table 4.2** — the block-length relief the earlier draft had to request is no longer needed — and no turnaround interval now approaches the 750-ft trigger of IFC 2024 (GA) Appendix D103.4. One thing got **worse**: the buffer, open-space and stream margins are all thinner than the earlier draft implied, and the required detention volume is now larger than the 2025 mixed-use filing's. Both are stated below at face value.

1. Property overview

1.1 Parcels

PIN (Gwinnett "R6123 ###")	Address	Deeded ac (2026 digest)	GIS-calculated ac	Owner of record (2026 digest)	2026 FMV (digest)
6123 033	4541 Arcade Rd SW	0.99	0.985	Awad, Mohammed	\$285,000
6123 015	4539 Arcade Rd SW	2.00	1.985	Awad, Mohammed	\$87,200
6123 014	4537 Arcade Rd SW	2.00	1.994	Mendez, Santos C. & Roblero de Leon, Lesvia Rosario	\$6,800
6123 162	4535 Arcade Rd SW	4.45	4.614	Mendez, Santos C. & Roblero de Leon, Lesvia Rosario	\$888,600
Assemblage		9.44	9.577	Land Lot 123, 6th District, Gwinnett County; City of Lilburn	\$1,267,600

- **Application acreage: 9.44 ac (deeded) = 411,206 SF.** The GIS parcel fabric computes **417,174 SF = 9.577 ac** (shoelace of the 55-vertex boundary ring); the RLS boundary survey governs. The 2025 filing's "9.00 ac" was incorrect (FACTS §1).

- **Current zoning:** R-1, Single-Family Residence (City of Lilburn), all four parcels. **Requested zoning:** R-2, Medium-Density Residential (Lilburn Zoning Ordinance 2023-603 §402).
- **Comprehensive Plan (adopted):** 2024 Comprehensive Plan character area **"Established Residential"** (printed p. 35, PDF p. 37); Ord. 2023-603 §203 compatibility table maps Established Residential → R-1, R-2, so the request is plan-consistent under the ordinance's own terms.
- **Comprehensive Plan Amendment (pending, not adopted).** The City Council held the transmittal public hearing on **2026-07-13** and voted to transmit the Amendment to the Georgia Department of Community Affairs. Under Ga. Comp. R. & Regs. 110-12-1 the DCA findings go to the Regional Commission within 35 days and the Regional Commission's report to the City within 40 days of submittal, so that window closed on or about **2026-08-23**. As of **2026-09-03 the Amendment has not been adopted**; adoption can be calendared at the **2026-09-14, 2026-10-12 or 2026-11-09** Council meetings — i.e. it may well be adopted law on or before the Council hearing on this case. The application is written so that it stands either way: the Amendment's Future Development Map places the site in class **"Suburban-Low"** (FDM p. 52), whose stoplight chart (p. 73) rates **Single-Family Cottages** and **Single-Family Appropriate** and Mixed Use (vertical and horizontal), Multifamily, Office, Retail/Entertainment, Hotel/Motel, Flex and Auto-oriented Retail *Not Appropriate*. The adjacent class, Suburban-Medium (p. 71), also rates both single-family categories *Appropriate*. **VERIFY with the Planning Director which class the four parcels fall in** — the FDM is a raster with no parcel labels — and the adoption date.
- **Shape:** a single strip **~236 ft wide at Arcade Rd** (234.7 ft measured NE-line to SW-line at $u = 0$; 236.1-ft frontage chord) **widening to ~246 ft at the NW end**, ~1,720–1,760 ft deep (SW line 1,757.01 ft; NE line 1,721.68 ft; rear line 246.41 ft). Only public frontage: Arcade Rd, 237.63 ft (docs/05 §5).
- **Prior zoning action:** RZ-2025-01 (R-1 → MU; 104 condominiums + 24,400 SF commercial) was tabled by the Planning Commission on 2025-08-28 for a defective legal advertisement, was never heard, and received no denial; the §1003-11 twelve-month bar is therefore not apparently triggered (formal disposition VERIFY in writing with the Planning Director).

1.2 Surroundings

Side	Local extent	Adjoining use	Subdivision (plat)	Zoning	Notes
SE (front)	$u \approx -36$ to 0	Arcade Rd, county-maintained 2-lane minor collector, 35 mph, 14,800 AADT (GDOT sta. 135-6689, 2017 — ineligible under Gwinnett DOT's two-year rule; see §6.2)	—	—	Arcadia Place centerline meets Arcade Rd at ($u -12$, $v +61$), 61 ft NE of the site's NE front corner
NE long side	$v > 0$	14 single-family lots: 4531 Arcade Rd (front corner) and 265–145 King David Dr, 0.45–0.76 ac each, 1970–71 brick split-levels/ranches, no HOA	King David Manor (plat S/159)	R-1 (Lilburn)	2026 digest FMV of abutting lots \$292,800–\$412,500, median \$349,200
SW long side	$v < -237$	14 single-family lots: 240–150 and 260, 250 Village Green Ct; 303 & 308 Fieldhouse Cir; 2007–2016 two-story homes 3,433–7,000 SF; HOA	Legends at Parkview (plat 118/187)	R-1 (Lilburn)	FMV of abutting lots \$579,100–\$894,600, median \$693,500
NW end (rear)	$u > 1,700$	3 single-family lots: 4520, 4530, 4540 Nantucket Dr, 1975–79	Nantucket (plats 1/268, 3/73, 4/157)	R-1 (Lilburn)	FMV \$308,400–\$417,000
Nearby precedent	~106 ft NE of the front corner, across 4531 Arcade	"Arcade Road Townhomes" R-2 polygon (Gwinnett zoning layer; case number VERIFY)	—	R-2 (Lilburn)	Nearest R-2; its sewer outfall crosses the subject site (\$2.5)
Nearby	~314 ft	Unincorporated Gwinnett R-100	—	R-100 (county)	—

Adjoining ownership: 31 privately owned tax parcels (14 NE, 14 SW, 3 NW), every one zoned R-1 in the City of Lilburn and developed with a single-family house, **plus the Gwinnett County Arcade Road right-of-way = 32 notice recipients**. The list is drawn from the Tax Assessor's 2026 ownership file (data/owners-found.json) and is reproduced at docs/06-adjoining-property-owners.md. Earlier drafts described "32 adjoining parcels ... developed with single-family houses"; the 32nd is a public road, not a house.

2. Existing conditions

2.1 Topography (USGS 3DEP 1-m DEM sampled on a 100 × 50-ft grid; NAVD88 ft; approximate — field survey required)

Location	Local u	Elevation (ft NAVD88)
Arcade Rd frontage	$u \approx 0$	≈ 940 (SW corner) – 950 (NE corner); lane centerline 943.5 at the entry join (u 149.5)

Location	Local u	Elevation (ft NAVD88)
Mid-swale	u ≈ 700–1,000	≈ 931–940 (lane centerline low point of the front reach 936.3 at u 779 ; 940.4 at u 979)
Ridge	u ≈ 1,080–1,200	952–958 (lane crest 953.5 at u 1,179)
Rear low pocket	u ≈ 1,350–1,500, SW side	921.6 at (u 1,384, v -217) — site minimum; lane low point 931.5 at u 1,479
NW end	u ≈ 1,700–1,760	943–948

Site range 921.6–958.2 ft; cross-slope generally falls NE → SW. Data: data/topo-samples.json; 2-ft contours in data/layout.json. **The steepest existing ground on the lane centerline is 10.9 % at u ≈ 1,494** — on the climb out of the rear low pocket — against the 12 % maximum street grade in the City's Site Development Plan Review Checklist. The finished profile is not yet designed and will need cut/fill and possibly retaining at the lot fronts through that reach (PE).

2.2 Streams and buffers

- An unnamed order-0 headwater channel (Gwinnett hydrology layer) enters the site at the SW line around u ≈ 1,356–1,392 and terminates inside the site at (u 1,391.7, v -209.8), about 30 ft inside the line. Path (u, v): (1262.8, -353.4) → (1341.7, -296.6) → (1356.7, -252.5) → (1364.9, -215.4) → (1391.7, -209.8).
- A second branch runs 38–58 ft outside the SW line at u ≈ 868–1,038, and a third reach is digitized off-site to the southwest. **All three reaches are now screened against every lot, dwelling, driveway, pond and pavement** (\$4.3 item 40) — the earlier revision screened only the on-site reach, and that omission is what removed a lot.
- Buffers applied:** 25-ft state buffer (O.C.G.A. §12-7-6(b)(15), measured from the point where vegetation has been wrested by normal stream flow); Lilburn Code Ch. 109 Art. VII (Stream Buffer Protection Ordinance, adopting the Metro North Georgia Water Planning District model text): **50-ft undisturbed buffer measured from the top of the stream bank, plus a 25-ft setback beyond it in which all impervious cover is prohibited = 75 ft**. The state and local lines are measured from **different data** and both are shown separately on the concept plan. (Article and section numbers within Ch. 109 Art. VII are to be transcribed from the City's copy — Municode refuses automated retrieval.)
- Two open questions the concept plan must not pretend to have settled.** (a) *Datum*. Every clearance computed to date is measured to the **digitized channel centerline**, not to top of bank; with a 3–8-ft channel half-width the true clearances are 3–8 ft smaller, so the design carries a **5.0-ft top-of-bank allowance** and screens every element at 75 + 5 = **80 ft** for impervious cover and 50 + 5 = 55 ft for land disturbance. (b) *Whether the buffer attaches here at all*. The ordinance defines a "stream" as beginning at a spring or seep that sustains streamflow, or at a point with **25 acres or more** of drainage area. The reach that enters this property drains far less than 25 acres, so the regulated stream may begin **downstream, off the property**. A field study — GA EPD's *Field Guide for Determining the Presence of State Waters That Require a Buffer* (Sept 2017) for the state half — must establish **whether and where** a stream exists before top of bank is located. The outcome could return the u 1,230–1,280 slot to lotting, or take more.
- Receiving water: Jackson Creek headwaters (GAR030701030315), on Georgia's 2024 303(d) list for bacteria and fish-community impairment (urban runoff) — water-quality treatment is treated as substantive (\$5.3).

2.3 Floodplain

FEMA **Zone X** (not a special flood hazard area) for the entire site per Gwinnett GIS (FACTS §2). No floodway or 100-year floodplain is mapped on the property; Table 4.2 "Floodways and Floodplains" is therefore not engaged. **VERIFY** against the effective FIRM panel (13135C0114F, eff. 2006-09-29 per docs/11 §2) at LDP.

2.4 Soils (NRCS SSURGO survey area GA135 via the Gwinnett GIS soils layer; names per docs/11-site-context-brief.md §3.3 — **VERIFY** against a current Web Soil Survey AOI report; the Soil Data Access endpoint was not reachable on 2026-08-29)

Map unit	NRCS name (VERIFY)	Slope	HSG	Depth to rock	Notable limitation	Where on site
AmC2	Appling sandy loam, 6–10 %, moderately eroded	6–10 %	B	none	Low-strength subgrade for roads	Front half, u 0–800 (lane and NE side)
ApB	Appling–Hard Labor complex, 2–6 %	2–6 %	B / C	none	Hard Labor: seasonal water table ≈ 32 in	Band at u ≈ 400–500
PfC2	Pacolet sandy loam, 6–10 %, moderately eroded	6–10 %	B	none	—	Ridge and NE flank, u 800–1,400
GeB2	Gwinnett clay loam, 2–6 %, eroded	2–6 %	B	paralithic 40–60 in	—	NW corner (u ≈ 1,700, SW)
GgE2	Gwinnett loam, 10–25 %, eroded	10–25 %	B	paralithic ≈ 50 in	Slope	Rear, u 1,400–1,700

Map unit	NRCS name (VERIFY)	Slope	HSG	Depth to rock	Notable limitation	Where on site
ARE	Ashlar–Rion–Waterree complex, 10–25 %	10–25 %	B	Ashlar: lithic bedrock 22–40 in	Slope, shallow rock	Mid-swale u 800–1,100 (lane and Pond 1) and u ≈ 1,500 SW (Pond 2)
HYB	Helena sandy loam, 2–6 %	2–6 %	D	none	Seasonal high water table ≈ 24 in; moderate shrink-swell	Front SW corner, u 0–200 (amenity block, entry, sewer tie-in)

No map unit on site is rated hydric by NRCS. Design consequences: **rock probes and an undercut allowance at both pond sites and the mid-swale lane before the detention volumes are trusted** (§5.3) — and that matters more now that both basins are 6 ft deep; moisture-conditioned fill and pavement-subgrade care on HYB at the entrance; slab-on-grade foundations suit the HSG B uplands.

2.5 Existing improvements and utilities

- **Structures: two older houses, and only one of them is near the road (corrected 2026-09-03).** **4541 Arcado Rd** (brick split-level, 1972; footprint ≈ 2,396 SF; 2026 digest heated area 2,053 SF) stands near the frontage, its footprint centred about (u 203, v –174) — inside the front amenity block, so it comes down in Phase 1. **4535 Arcado Rd** (1987; footprint ≈ 4,568 SF; digest heated area 4,267 SF with a finished basement) does **not** stand near the frontage: it sits about **1,200 ft back from the right-of-way at about (u 1,202, v –114), on the u 1,080–1,200 ridge and directly on the proposed lane alignment**. It is therefore a **Phase-2 demolition standing on the lane**, not a frontage structure, and no Phase-2 land disturbance can begin until it is removed. Earlier revisions of this summary said "two older houses ... near the front"; that was wrong. Accessory structures and both existing driveways are likewise removed. 4535's septic-vs-sewer status is VERIFY. Footprints are affine-fitted from public aerial imagery by `tools/existing_conditions.py` and are approximate — the RLS survey locates them. Existing structures and the tree line are drawn on Sheet C-1.0 (Existing Conditions).
- **Sanitary sewer (existing):** an 8-in gravity main, Gwinnett DWR project "Arcado Road Townhomes Sewer Outfall", runs from Arcado Rd along the SW edge of parcel 033 (v ≈ –224 to –228, 7–13 ft inside the line) from u ≈ –42 to u ≈ 272 — a run of about **314 ft on the property** — then exits southwest toward King David Manor (the "EX. 20' SSE" on the 2025 hydro sheet). Inverts (ft): 943.79 / 940.47 at Arcado → 932.69 / 930.84 (u ≈ –42 → 20) → 930.64 / **927.13 at the manhole near (u 272, v –224)**, i.e. about **272 ft in from Arcado Rd** → 925.17 downstream. A downstream 8-in main in Legends at Parkview has invert **919.58** at the manhole near (u ≈ 1,349, v ≈ –405). Inverts are GIS attributes — **invert survey required**.
- **Water:** DIP/ACP mains in Arcado Rd 47–107 ft from the front line; nearest hydrant ≈ 61 ft from the front corner (u –45, v +41).
- **Fire service:** Gwinnett County Fire & Emergency Services (the city has no fire department); first-due Station 3 ≈ 2.9 mi.
- **Environmental history:** no GA EPD Hazardous Site Inventory or UST records found for the parcels (brief §6.1); no industrial legacy.

3. Proposed development

3.1 Program

Item	Proposed	Basis
Use	Single-family detached cottage homes on fee-simple lots (Ord. 2023-603 §602 Use Table: "Single-family (cluster-cottage, creative lot configuration)", P in R-2, no SUP)	FACTS §3, §4
Age restriction	55+ under the Housing for Older Persons Act (42 U.S.C. §3607(b)(2)(C); 24 CFR §100.300–.308) — at least 80 % of occupied units with at least one occupant 55 or older. At 41 units the computation is $0.80 \times 41 = 32.8$, and 24 CFR §100.305(g) counts the fraction into the 80 percent, so 33 of the 41 units must have a 55+ occupant, leaving at most 8 that need not. Offered as a voluntary zoning condition and a recorded covenant	24 CFR §100.305(a), (g)
Dwelling units	41 (14 SW side, 27 NE side). FACTS §4's "≈44 lots, acceptable range 40–47" is a <i>target</i> , not a result; 41 is what the corrected concept plan draws, and it is still inside that range	data/layout.json
Density	4.34 du/ac on 9.44 ac deeded (4.28 du/ac on 9.577 ac GIS) vs. 8 du/ac R-2 maximum	$41 \div 9.44$; $41 \div 9.577$
Stories / height	1 story; ridge ≤ 24 ft offered as a condition (40 ft permitted by Table 4.1). As drawn on Sheets A-2.1 / A-2.2: Plan A 17'-10" and Plan B 19'-0" above finished floor = 18'-6" and 19'-8" above finished grade (finished floor is 8 in above grade at the front). data/plans.json still computes Plan B's main gable at a 29.5-ft span, i.e. 19'-4" (19.33 ft) above the floor — a 4-in difference to be conformed in the data file, not in the drawing (§10 item 4)	Sheets A-2.1 / A-2.2; data/plans.json
Tenure / maintenance	Fee-simple lots; mandatory HOA maintains the lane, common tracts, yards, ponds and buffers	FACTS §4
Not proposed	No commercial, attached units, mixed use, daycare, multifamily, or conservation-subdivision SUP. No wall signs on any structure; one monument entry sign	FACTS §4

3.2 Lot statistics (recomputed from the `data/layout.json` polygons, then cross-checked against the generator's summary block)

Statistic	Value
Lots	41 — 14 on the SW side, 27 on the NE side
Typical lot	50'-0" × 100'-0" = 5,000 SF (Table 4.1 R-2 cottage minimum 3,000 SF; detached-SF minimum 5,000 SF)
Lot area min / avg / max	5,000 / 5,000 / 5,000 SF (shoelace of every lot polygon)
Lot width min	50.00 ft every lot (Table 4.1 minimum 50 ft)
Lot depth min (perpendicular, front lot line to rear lot line)	100.00 ft every lot (Table 4.1 minimum 100 ft). The 0.006-ft shortfall reported in the earlier draft was a rounding artifact and is gone from the regenerated geometry
Lot frontage on the lane tract	50.00 ft every lot (\$319 minimum 30 ft)
Where the lots are	SW side u 230–1,280 (14 lots, with breaks at the u 530–580 green, the Pond 1 tract at u 780–980, the u 1,080–1,130 green and the u 1,230–1,280 stream-setback tract); NE side u 230–1,730 (27 lots, continuous but for the two greens). No lot fronts Arcade Rd — the nearest lot line is 230.0 ft from the right-of-way
Total lot area	205,000 SF (4.71 ac; 49.1 % of the GIS site)
Rear 20 ft of every lot	20-ft undisturbed buffer held in a recorded buffer/conservation easement — $41 \times 50 \text{ ft} \times 20 \text{ ft} = \mathbf{41,000 \text{ SF}}$, reported separately from common open space
Setback envelope per lot	front 15'-0" minimum to the porch face (drawn at 15'-8"), sides 5'-0" minimum (drawn at 6'-0"), rear 20'-0" (= the buffer easement); garage door 26'-8" from the front lot line , so the driveway holds two cars clear of the sidewalk
Plan mix	21 × Plan A, 20 × Plan B (alternating)
Phase 1 (u < 980) / Phase 2	24 lots / 17 lots

3.3 The two plans (DRAFT — the architect's sealed plans supersede; see Sheets A-1.1/A-2.1 (Plan A), A-1.2/A-2.2 (Plan B) and A-2.3 (colours and materials) in `drawings/`)

Areas below are read from `data/plans.json` — the file the architectural sheets are drawn from — and supersede the "38'-0" × 38'-0" / 40'-0" × 40'-0"" nominal bodies in FACTS §4. Both plans are **L-shaped**: the 2-car garage has to be a front wing to keep the body inside the 40'-0" buildable width (50-ft lot less two 5-ft side yards).

	Plan A "The Springbrook"	Plan B "The Laurel"
Overall body (outside face, conditioned + garage; porch and patio excluded)	38'-0" × 51'-10"	38'-0" × 57'-6"
Program	2 BR / 2 BA	2 BR + den / 2 BA, fireplace, walk-in pantry, laundry/mud room off the garage
Conditioned (heated) floor area	1,406 SF (program target 1,444 SF)	1,492 SF (program target 1,520 SF)
Garage (2-car, front-loaded, recessed 5'-0" behind the front wall)	$21'-0" \times 21'-10" = \mathbf{458 \text{ SF}}$	$21'-0" \times 24'-0" = \mathbf{504 \text{ SF}}$
Covered front porch	$17'-0" \times 6'-0" = \mathbf{102 \text{ SF}}$	$17'-0" \times 6'-0" = \mathbf{102 \text{ SF}}$
Rear porch / patio	uncovered patio $12'-0" \times 6'-0" = \mathbf{72 \text{ SF}}$ (no covered rear porch)	covered rear porch $14'-0" \times 6'-0" = \mathbf{84 \text{ SF}}$, recessed under the main roof
Total under roof (conditioned + garage + covered porches)	1,966 SF	2,182 SF
Building coverage on a 5,000-SF lot (under roof, before driveway)	39.3 %	43.6 %
Maximum ridge (as drawn on Sheets A-2.1 / A-2.2)	17'-10" above finished floor = 18'-6" above finished grade	19'-0" above finished floor = 19'-8" above finished grade
Rear structure to the rear lot line	patio edge 20'-6"	rear porch edge 20'-10"
Side yard as sited	6'-0" both sides	6'-0" both sides
Accessibility	zero-step entry, 36-in doors, 5-ft turning circles in baths, blocking for grab bars (voluntarily exceeds §734, which applies to attached dwellings only)	same

	Plan A "The Springbrook"	Plan B "The Laurel"
Exterior	fiber-cement lap and board-and-batten; brick or stone water table and porch piers; architectural-shingle roof; 8:12 main and front-wing gables, the front-wing gable carried 6 ft forward to roof the covered porch (there is no shed porch roof), and a subordinate 6:12 gable over the garage ; window sills at 2'-8"; three colour schemes (warm white / sage / clay)	same

- **Total conditioned area: $21 \times 1,406 + 20 \times 1,492 = 59,368$ SF, averaging 1,448 SF per unit.** Both plans clear the Table 4.1 minimum heated floor area of 1,000 SF for a cottage home with room to spare.
- The garage is a separate front wing, computed independently and **excluded** from conditioned area. Earlier drafts offered staff an alternative reading in which the garage sat inside the body ($\approx 1,024 / \approx 1,180$ SF heated); the filed drawings disprove it and it is withdrawn.
- Under the 2026 Comprehensive Plan Amendment's own vocabulary (land-use table p. 54) a "Single-Family Cottage" is a house "typically 1,200 square feet or less" and "Single-Family" is one "typically exceeding 1,200 square feet". At 1,406 and 1,492 SF these are **"Single-Family"** in the plan's terms — a category the Suburban-Low chart also rates *Appropriate*.
- **One drawing-versus-data difference is open on Plan B's ridge.** The issued elevation Sheet A-2.2 prints **19'-8" above finished grade** (19'-0" above the floor). data/plans.json computes the same gable at a 29.5-ft span and prints **19'-4" (19.33 ft) above the floor**, which would be 20'-0" above grade. The difference is 4 in and is below the 24-ft voluntary cap either way, but a recorded condition must not cite a dimension the filed sheet does not draw, so this document and voluntary condition 3 state the **drawn** figures and the data file is the one to conform (§10 item 4; docs/09 §5.3 item G-1). **VERIFY** with the architect.

3.4 Amenity block (front, u ≈ -36 to 230; common tract 46,777 SF)

- Clubhouse **2,400 SF (40'-0" $\times$ 60'-0")**, one story, **114.8 ft** (true perpendicular) from the Arcado Rd right-of-way — well beyond the 50-ft collector setback of Table 4.1 — with a village green.
- **2 pickleball courts** (20'-0" $\times$ 44'-0" courts on 30'-0" $\times$ 60'-0" fenced pads), set back from the R-1 lines; no court lighting (voluntary condition 11).
- Mail kiosk (USPS cluster-box unit) with **4 short-term spaces**; **8 guest spaces** (7 standard + 1 van-accessible) in a small bay. **Both bays now lie inside the lane tract, not the amenity tract:** the tract widens to 60 ft through the amenity block (u 149.5–230) so that the 18-ft-deep 90° bays sit wholly within the tract that owns and maintains them. Their 2,070 SF is therefore counted as lane pavement in §3.7, not as amenity paving.
- 10-ft landscape strip along the Arcado Rd right-of-way; one monument entry sign ≤ 32 SF and ≤ 6 ft high, standing **19.0 ft** from the right-of-way — inside the 50-ft principal-structure setback, as monument signs normally are, shown on the plan as an express exception and to be confirmed against the Lilburn sign regulations (**VERIFY**); a 5-ft sidewalk along the Arcado Rd frontage.
- **6,760 SF of the amenity tract is building or pavement** (clubhouse roof 2,400 + two court pads 3,600 + mail kiosk 160 + an amenity-walk allowance 600). Added to the **7,200 SF** of hammerhead turnaround legs that lie inside the pocket and terminus greens, **13,960 SF of the common open space is built or paved**. That is disclosed here and in §3.7 rather than left for a reviewer to discover inside the open-space number.

3.5 Access and private lane

Item	Value	Basis
Access	One full-movement driveway on Arcado Rd; no driveway from any lot to Arcado Rd; no gate. Gwinnett UDO §900-40.5(C) allows one driveway per 400 ft of frontage — at 237.63 ft of frontage, one is the maximum, which independently supports the single-entrance design	FACTS §4; Gwinnett UDO §900-40.5(C)
Entrance location	SW third of the frontage, centered at v = -190 , throat at u = -31.4 ; the entry drive curves to the lane midline and joins it at u 149.5 . Separation from the Arcadia Place centerline (v +61): 251.2 ft along the Arcado Rd centerline (250.7 ft chord) against the 244-ft minimum of Gwinnett UDO §900-40.5, Table 900.3 at 35 mph — 7.2 ft of margin	layout.json lane.entrance
Entrance geometry	24'-0" drive; curb returns R = 25'-0" northeast, R = 15'-0" southwest ; two reverse curves R = 100'-0" through a 59.9° deflection; drive length 204.0 ft (AASHTO minimum radius at 15 mph is 47 ft, so 100 ft is generous)	layout.json
Buffer at the SW curb return — a reduction is requested	Even with the southwest return cut from 25'-0" to 15'-0" , its tangent point on the right-of-way lies 2.0 ft inside the 20-ft perimeter buffer band, against 303 Fieldhouse Cir SW — so the buffer narrows to a minimum of 18'-0" instead of 20'-0" at that one corner, and the pavement actually inside the band is 0.1 SF . The entrance cannot be shifted northeast without spending the 7.2 ft of driveway-spacing margin. This is a buffer reduction; the Letter of Intent §5 requests it and voluntary condition 4 states it , as 2026 Application Instructions item 6 and Ord. 2023-603 §1003-4.5 require, and it is dimensioned on the entry enlargement sheet. The buffer is at its full 20 ft everywhere else on the SW, NE and NW lines	layout.json metrics sw_curb_return_buffer_encroachment_ft / _sf; audit SG-07; data/voluntary-condition s.md 4

Item	Value	Basis
Lane	Private, HOA-owned tract, 1,754 ft of travelled way from the Arcade Rd right-of-way to the end of pavement at u 1,700 (entry drive 204.0 ft + lane centreline 1,550.5 ft = 1,754.5 ft), dead-end; 22'-0" pavement , 2-ft strip, 5-ft sidewalk on the NE side for the whole length and on the entry drive, second 5-ft sidewalk on the SW side from u = 530; tract width 35.9 ft minimum to 46.1 ft maximum , widening to 60 ft through the amenity block; design speed 15 mph	layout . json
Lane and sidewalk areas	pavement 48,430 SF (lane 34,111 + entry drive 4,865 + hammerhead legs 7,200 + guest/kiosk bays 2,070 + curb returns 184); sidewalk 14,619 SF ; lane tract (lane + entry drive) 71,085 SF	layout . json; recomputed
Turnarounds	3 hammerhead (T) turnarounds, 20'-0" wide × 60'-0" legs, at u 540, 1,110 and 1,690 (terminus) — stations 594.5 / 1,164.5 / 1,744.5 ft from the right-of-way, spacing 594.5 / 570.0 / 580.0 ft. No circular cul-de-sac. Every interval is now well inside the 750-ft turnaround interval of IFC 2024 (GA) Appendix D103.4 — the earlier layout's 746.4-ft first interval, which cleared it by 3.6 ft, is gone	layout . json; audit SG-08
Blocks	Read between street intersections (the Arcade Rd right-of-way and the three hammerhead T-legs) the runs are 594.5 / 570.0 / 580.0 ft ; read as lot-frontage runs between the open-space breaks they are 300 / 500 / 550 ft. Table 4.2 asks for block lengths not exceeding 600 ft , and both readings are now inside it — the longest is 594 ft. A pocket green is an open-space tract, not a street, and cannot terminate a block, so the first reading is the honest one; it no longer needs relief	layout . json metrics.blocks_ft; audit SG-06
Lane half-section	The checklist section is 22'-0" of pavement plus, on each walked side, a 2-ft strip and a 5-ft walk: 29.0 ft for one walk, 36.0 ft for two . The tract is the residual after two 100-ft lot depths — 35.9 ft at the narrowest row , which carries the pavement and one walk with 6.9 ft to spare but is 0.1 ft short of two. The SW walk therefore starts at u 530, where the strip has widened; over u 230–530 the SW half-width is 16.94–18.31 ft	layout . json; \$4.3 item 36
Emergency access	No secondary access is physically possible — no street abuts the rear or either long side (Nantucket Dr lots close the NW end). Mitigation per \$5.4	FACTS \$4

3.6 Parking

Item	Count	Basis
Required — Table 8.1, single-family detached, 2 per DU	$41 \times 2 = \mathbf{82}$	Ord. 2023-603 Table 8.1
Provided on lots — 2 garage + 2 driveway per DU	$41 \times 4 = \mathbf{164}$	driveway 20'-0" wide × 26'-8" to the garage door
Guest spaces (bay at the amenity block)	8 (7 standard + 1 van-accessible with an 8-ft aisle)	layout . json
Mail-kiosk short-term spaces	4	layout . json
Total	176 (1 guest/kiosk space per 3.4 lots; FACTS target 1 per 4)	

3.7 Open space, impervious cover and site areas

Common open-space tract	Area (SF)
Front amenity tract (clubhouse, village green, courts, landscape strip)	46,777
Pocket green, u 530–580 (both sides of the lane)	10,000
Pocket green, u 1,080–1,130	10,000
Terminus green + rear buffer	9,312
Pond 1 tract (dry detention / water quality), SW side u 780–980	20,000
Pond 2 tract (dry detention / water quality), SW side u 1,480–1,680	20,000
Creek woods — stream head with 50/75-ft buffers, preserved	20,000
Stream-setback tract, SW side u 1,230–1,280 — unlotted, no impervious cover	5,000
Total common open space (enumerated)	141,089 SF = 3.24 ac = 33.8 % of the 417,174-SF GIS area (34.3 % of 9.44 deeded ac)

- **One number, enumerated, and it now closes exactly.** The tract-by-tract sum equals the residual (boundary – lots – lane tract) to the square foot: 417,174 – 205,000 – 71,085 = **141,089 SF**. The earlier draft carried 132,959 SF against a 133,159-SF residual with 200 SF of unallocated slivers; there are none now.
- **Of that total, 13,960 SF is building or pavement** — 6,760 SF in the amenity tract and 7,200 SF of hammerhead turnaround legs inside the greens (\$3.4). **Green-only open space is 127,129 SF = 30.5 % of the GIS area.**
- The **41,000 SF** of 20-ft buffer easement on the lots is **not** counted in the 33.8 %.
- **Open space is not a code requirement here.** Table 4.1 refers open space to Development Regulations §5.9, which has been retrieved: §5.9.1 "Recreation Areas" applies to "single-family detached subdivisions having a gross area of **50 acres or more**",

to duplex subdivisions of 50 acres or more, and to attached, townhouse and apartment developments of ten acres or more. At 9.44 acres of single-family detached, **\$5.9 is not triggered**, and the 33.8 % of common open space is entirely voluntary. (§5.9's text still names "Gwinnett County," "the Board of Commissioners" and "the Department" because Lilburn adopted the county regulations wholesale — confirm with staff who those actors are in a city case.)

- **Site-area accounting (GIS 417,174 SF):** lots 205,000 + lane tract 71,085 + common open space 141,089 = 417,174 SF, with nothing left over.
- **Impervious cover: 184,148 SF = 4.23 ac = 44.1 % of the GIS site (44.8 % of the deeded acreage).**

Impervious component	SF	Basis
Dwellings, garages, porches, rear porches and patios	86,448	21 × (1,966.5 under roof + 72 patio) + 20 × 2,182, from the data/plans.json footprints
Lot driveways and entry walks	31,390	the 41 drawn driveways (653–764 SF each, 706 average) + a 60-SF entry walk per lot
Private lane pavement	34,111	22'-0" × 1,550.5 ft
Entry drive pavement	4,865	the drawn polygon: 24'-0" wide on a 204.0-ft centreline (31 SF under the 4,896-SF nominal product, the arcs being drawn as chords)
Entrance curb returns	184	the two drawn fillets, R = 25 ft NE and R = 15 ft SW
Hammerhead turnaround legs	7,200	3 turnarounds × 2 legs × 20'-0" × 60'-0"
Guest and mail-kiosk parking bays	2,070	63 × 18 + 52 × 18, inside the lane tract
Sidewalks	14,619	shoelace of the three drawn walks
Clubhouse roof, court pads, mail kiosk and amenity walks	6,760	\$3.4
Less driveway apron / sidewalk overlap counted twice	–3,500	35 crossings × 20 ft × 5 ft
Total	184,148	44.1 % of GIS / 44.8 % of deeded

- The components as printed sum to 184,147 SF; the 1-SF difference from the stated total is rounding of the individual polygons. Independently, tools/dev_summary_calcs.py rebuilds the same quantity from the lot polygons and data/plans.json and gets **184,178 SF** — 0.02 % apart, the difference being how the curb-return fillets and the bay outlines are allowed for. Either figure rounds to 44.1 %.
- This supersedes the 160,167 SF / 38.4 % and 161,212 SF / 38.6 % figures carried in earlier drafts, both of which rest on a hard-coded allowance of 2,200 SF of impervious per lot. The real houses are larger than that allowance: **2,752 SF per Plan A lot and 2,898 SF per Plan B lot**, roof plus patio plus driveway plus walk.
- **Against the 2025 mixed-use filing** (hydrology sheet HYD-1: 235,388 SF impervious, 55.4 %), the cottage plan reduces impervious area by **21.8 %** — not the "–32 %" earlier drafts claimed on the understated figure.

3.8 Phasing

- **Phase 1** — $u < 980$: entrance, amenity block, lane to and beyond the first hammerhead turnaround (u 540, station 594.5 ft), Pond 1 (tract u 780–980), **24 lots**, gravity sewer to the existing on-site manhole. Phase 1 needs **no sewer extension of any kind**, and the only demolition it requires is 4541 Arcade Rd.
- **Phase 2** — $u \geq 980$: remaining lane, Pond 2, creek-woods and stream-setback preservation, **17 lots**, after the rear sewer route (§5.2) is settled with Gwinnett DWR. It also requires the demolition of **4535 Arcade Rd, which stands on the lane alignment at $u \approx 1,202$** (§2.5). The first turnaround and the tract layout allow Phase 1 to stand alone.

4. Required-vs-provided matrix

Status vocabulary: **appears consistent** / **interpretation requested** / **relief requested** / **VERIFY**. Citations are to the Lilburn Zoning Ordinance, Ord. 2023-603, unless noted. "Provided" is the current data/layout.json / data/plans.json geometry (regenerated 2026-09-03).

4.1 Use and dimensional standards

#	Requirement (citation)	Required (R-2)	Provided	Status
1	Use — "Single-family (cluster-cottage, creative lot configuration)" (§602 Use Table)	P — permitted, no SUP	41 detached cottage homes on fee-simple lots; no conservation-subdivision SUP, no multifamily	appears consistent
2	District intent (§402-1)	"detached single-family ... clustered and cottage housing types ... transition between lower density single-family districts and the MU District"	1-story detached cottages at 4.34 du/ac between two R-1 subdivisions	appears consistent
3	Minimum lot size, cottage home (Table 4.1)	3,000 SF	5,000 SF minimum, average and maximum (also meets the 5,000-SF detached-SF row)	appears consistent
4	Minimum lot width, cottage home (Table 4.1)	50 ft	50'-0" every lot	appears consistent
5	Minimum lot depth, all uses (Table 4.1)	100 ft	100.00 ft every lot	appears consistent
6	Maximum gross density (Table 4.1)	8 du/ac	4.34 du/ac (9.44 deeded ac); 4.28 du/ac (GIS)	appears consistent
7	Minimum heated floor area, cottage home (Table 4.1)	1,000 SF	Plan A 1,406 SF , Plan B 1,492 SF conditioned (garage excluded)	appears consistent
8	Maximum height (Table 4.1)	40 ft	1 story; ridge 17'-10" (A) and 19'-0" (B) above finished floor = 18'-6" and 19'-8" above finished grade as drawn; 24-ft cap offered as a condition	appears consistent
9	Buffer abutting R-1 property (Table 4.1)	0 ft for "detached single-family dwellings in R-2"; 20 ft for "all other allowed dwelling types"	20-ft undisturbed buffer along the SW, NE and NW lines (the stricter row), in recorded easements on the lots and within common tracts elsewhere; one 2.0-ft reduction requested at the SW curb return (item 41a)	interpretation requested (Q1) — is a detached cottage home a "detached single-family dwelling" (0 ft) or an "other allowed dwelling type" (20 ft)? The plan provides 20 ft either way
10	Buffers supersede minimum required yards (§313(1))	"Buffer requirements ... supersede these minimum required yards"	The 20-ft buffer coincides with the 20-ft rear yard of each perimeter lot. At the narrowest lot row the strip is 235.9 ft wide and the section is 100 + 35.9 + 100 = 235.9 ft — the lane tract is the residual and the arithmetic closes with nothing to spare	interpretation requested (Q2) — see the caution below. There is no slack: every foot the boundary survey removes from the width comes straight off the lane tract
10a	Fallback if Q2 is answered "separate tract required"	Lot depth 100 ft (Table 4.1)	Lots 50'-0" × 82'-0" (4,100 SF > 3,000 SF) with the 20-ft buffer in a separate tract, and a concurrent \$1005 variance on lot depth (criteria §1005-3.2 a–g; exceptional narrowness of a 236-ft-wide, 1,740-ft-deep strip as the hardship). A third, shallower house plan is required — the buildable depth on an 82-ft lot is 82 – 15 – 6 – 20 = 41'-0", and neither Plan A (51'-10") nor Plan B (57'-6") fits	relief requested (conditional fallback only)
11	Front setback, local street (Table 4.1)	15 ft from the right-of-way	15'-8" from the private-lane tract line to the porch face; the 8-in roof overhang is held outside the required yard	appears consistent — VERIFY that staff measures from the private-tract line
12	Front setback, collector/state road (Table 4.1)	50 ft from the Arcado Rd right-of-way	No lot fronts Arcado Rd; nearest lot line 230.0 ft ; nearest principal structure (clubhouse) 114.8 ft , true perpendicular. The monument entry sign stands 19.0 ft from the right-of-way and is not a principal structure — shown as an express exception (item 41b)	appears consistent — sign setback VERIFY
13	Side setback (Table 4.1)	5 ft	6'-0" both plans as sited (data/plans.json lot_siting.side_yard_ft = 6.0 for Plan A and Plan B alike)	appears consistent
14	Side setback, collector/state road (Table 4.1)	20 ft from the right-of-way	No lot has a side on Arcado Rd	not applicable

#	Requirement (citation)	Required (R-2)	Provided	Status
15	Rear setback (Table 4.1)	20 ft	Plan A patio edge 20'-6" ; Plan B rear porch edge 20'-10" ; no structure inside the rear 20-ft buffer easement	appears consistent — the clearance is 6 in and 10 in, and is to be re-checked when the architect seals the plans
16	Accessory building setbacks (Table 4.1)	5 ft side / 5 ft rear	No accessory buildings proposed on lots (HOA covenant)	appears consistent
17	Open space (Table 4.1 → Development Regulations §5.9, App. B)	§5.9.1 "Recreation Areas" applies only to single-family detached subdivisions of 50 acres or more	9.44 ac — §5.9 is not triggered . 141,089 SF = 33.8 % of the GIS area is provided voluntarily	resolved — not applicable ; provided voluntarily
18	Lot-size averaging / width reduction / setback adjustments (§309, §310, §314)	Available only in an Overlay District or redevelopment Character Area	Not used; not available here	not applicable
19	Street frontage (§319)	Each lot abuts ≥ 30 ft on a public street or an approved private street	50'-0" on the private lane; lane built to public-street standards per the Site Development Plan Review Checklist	appears consistent — VERIFY the lane's approval as an "approved private street" (Q4)
20	Floodplain (§325; Table 4.2)	No development in floodways; restrict development in floodplains	FEMA Zone X; no floodplain on site	appears consistent

Caution on item 10 — the reading that carries the lot yield. §313(1) says "Buffer requirements established by this article ... supersede these minimum required yards." "This article" is Article 3, and Article 3 sets no buffer widths (§318 imports the district buffers of Table 4.1 by reference). Nothing in the ordinance expressly authorizes holding a required buffer **inside a private lot** under an easement. The applicant therefore presents §313(1) as the **design basis and an interpretation to be confirmed**, not as settled law, and carries the item 10a fallback in the open. This is the single most consequential question at the pre-application conference.

4.2 Table 4.2 design criteria (R-2 column "A" = applicable)

#	Criterion (Table 4.2)	Provided	Status
21	Compatibility with existing character	1-story cottages, ridges 18'-6" and 19'-8" above grade — the lowest-massed buildings on the strip; fiber-cement and brick/stone; 20-ft undisturbed buffers on all three shared lines	appears consistent
22	Floodways and floodplains	None on site (Zone X)	appears consistent
23	Access to abutting public conservation lands	No public open space or greenway abuts the site	not applicable
24	Road and lot layout — grid, rectangular lots, block widths and lengths not exceeding 600 feet	Rectangular 50'-0" × 100'-0" lots; a single-spine lane is dictated by a 236-ft-wide strip with one frontage. Measured between street intersections the runs are 594.5 / 570.0 / 580.0 ft ; measured as lot-frontage runs between the open-space breaks, 300 / 500 / 550 ft. The longest block on either reading is 594 ft , inside the 600 ft the table asks for	appears consistent — relief no longer requested. The earlier layout's 746-ft first run needed a waiver; re-spacing the turnarounds removed the need
25	Culs-de-sac — "Circular turnarounds shall not be permitted except in the most unusual circumstances"	No circular turnaround anywhere; 3 hammerhead (T) turnarounds; the terminus is a T within a green	appears consistent
26	Non-pervious surfacing — pervious materials allowed	Pervious pavers proposed for the guest bay and mail-kiosk spaces; the lane in asphalt for fire-apparatus loading (PE to confirm)	appears consistent
27	Parking between the front façade and the street discouraged	Garages recessed 5'-0" behind the front wall and the porch projects 6 ft beyond that; driveway pairs are the only parking between façade and lane (needed for the two on-lot spaces beyond the garage); guest parking is off the lane in a bay	appears consistent — noted for staff
28	Alley access to parking encouraged / may be required	Not feasible: a rear alley would fall in the 20-ft buffer on a 236-ft-wide site	interpretation requested (waiver of the "may be required" clause)
29	Front-loaded garages recessed ≥ 5 ft behind the front wall	5'-0" recess on both plans, every lot	appears consistent

#	Criterion (Table 4.2)	Provided	Status
30	Land consumption minimized; perimeter buffers and open space preserved	5,000-SF lots (against 9,000 SF in R-1); 33.8 % common open space; continuous 20-ft perimeter buffer	appears consistent
31	Open-space design / landscape ecology	Creek woods preserved with the 50/75-ft buffers; the u 1,230–1,280 stream-setback tract left unlotted; ponds in the swale and the rear low pocket; greens on the lane	appears consistent
32	Greenbelts and wildlife corridors	Continuous 20-ft buffer on three sides plus the creek-woods and stream-setback tracts forms a perimeter greenbelt	appears consistent
33	Conservation principles for major conservation subdivisions	Not a conservation subdivision (§706 SUP not sought)	not applicable

4.3 Parking, streets, buffers, fire and other standards

#	Requirement (citation)	Required	Provided	Status
34	Parking, single-family detached (Table 8.1)	2 per DU → 82	164 on lots + 8 guest + 4 kiosk = 176	appears consistent
35	Interior drive width (Site Development Plan Review Checklist)	20 ft two-way, no parking; 22 ft with 90° parking	22'-0" pavement throughout, no on-lane parking; the tract widens to 60 ft through the amenity block so the 18-ft-deep 90° bays lie inside it	appears consistent
36	Sidewalks (checklist)	5 ft on all roads, 2 ft off back of curb (6 ft with street trees)	5-ft sidewalk on the NE side of the whole lane and on the entry drive; second 5-ft sidewalk on the SW side from u 530; 5-ft sidewalk along the Arcade Rd frontage	partial — VERIFY. Two walks need a 36.0-ft tract and the narrowest is 35.9 ft, so 6 of the 14 SW lots (u 230–530) front a reach with no walk on their side. Closing it costs either 0.5 ft of lot depth or a 1.5-ft strip. Ask at pre-application and record the answer
37	Landscape strip along a public right-of-way (checklist)	10 ft	10 ft along Arcade Rd, planted at 1 tree and 1 shrub per 25 lineal ft → 10 trees and 10 shrubs over 237.63 ft	appears consistent
38	Cul-de-sac / dead-end length (checklist)	≤ 2,000 ft	1,754 ft	appears consistent (fire-code review is separate — item 43)
39	Street grades (checklist)	≤ 12 %	Existing ground on the lane centerline reaches 10.9 % at u ≈ 1,494 ; the finished profile is not yet designed	VERIFY at engineering — a PE plan-and-profile is required before the section is claimed of the finished lane
40	Stream buffers (Lilburn Code Ch. 109 Art. VII, 50 ft + 25 ft; O.C.G.A. §12-7-6(b)(15), 25 ft)	Undisturbed 50 ft from top of bank; no impervious cover within 75 ft	All three digitized reaches are screened, not only the on-site one , and every element is held at ≥ 80 ft from a digitized centerline — 75 ft plus the 5.0-ft top-of-bank allowance: nearest lot 80.6 ft , nearest dwelling 106.2 ft , nearest driveway 102.6 ft , nearest pond top of bank 80.2 ft , lane pavement 80.5 ft . Worst impervious clearance 80.2 ft, or 75.2 ft net of the allowance. The SW slot at u 1,230–1,280 that failed the screen is not lotted ; it is a stream-setback open-space tract	appears consistent — top of bank to be field delineated ; the 5.0-ft allowance is a design assumption, not a survey. This screen is what took the yield from 43 to 41
41	Buffer encroachment by ponds and utilities (checklist)	≤ 50-ft width perpendicular; parking ≥ 5 ft from buffers	No pond now encroaches the perimeter buffer : Pond 1's top of bank is 41.5 ft and Pond 2's is 30.0 ft from the SW line. The guest bay is > 5 ft from any buffer	appears consistent — the earlier layout's 0.1-ft Pond 2 encroachment is gone
41a	Perimeter buffer at the entrance (Table 4.1; §1003-4 buffer reduction)	20-ft undisturbed buffer abutting R-1	The SW curb return is reduced from 25'-0" to 15'-0"; 2.0 ft measured along the right-of-way, and 0.1 SF of pavement, remain inside the band at the right-of-way corner — the buffer narrows to 18'-0" there. The entrance cannot be shifted NE: it is already only 251 ft from the Arcadia PI centerline against the ~244-ft spacing standard	relief requested — a buffer reduction of 2.0 ft over 0.1 SF is requested in the Letter of Intent §5, stated in voluntary condition 4 and dimensioned on the entry enlargement sheet

#	Requirement (citation)	Required	Provided	Status
41b	Monument entry sign in the 50-ft collector setback	50-ft principal-structure setback (Table 4.1)	One monument sign ≤ 32 SF, ≤ 6 ft high, 19.0 ft from the right-of-way; a sign is not a principal structure	VERIFY against the Lilburn sign regulations; shown as an express exception on the plan
42	Tree density (Code Ch. 109 Art. V, §109-168(3))	16 TDU/ac exclusive of buffer acreage	Tree survey and preservation plan at LDP; preserved creek woods, stream-setback tract and buffers credited. The Site Development Plan Review Checklist §6.i does not require TDU calculations for a single-family-residential subdivision	VERIFY at LDP
43	Dead-end fire apparatus road (IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04 , Appendix D103.4)	Table D103.4: over 750 ft → "Special approval required" ; the 501–750-ft band asks 26 ft of width	1,754-ft single access; 22'-0" clear lane; 3 hammerhead turnarounds at 594.5 / 570.0 / 580.0-ft intervals, every one inside the 750-ft trigger	special approval requested — Gwinnett County Fire Marshal. The 22-ft width against the table's 26 ft is the relief item
44	Two access roads for one- and two-family developments (Appendix D107.1 as modified in Georgia)	Ga. Comp. R. & Regs. 120-3-3-.04 replaces the model IFC threshold of 30 with 120 : two access roads are required where the number of dwelling units "exceeds 120"	41 units — the requirement is not triggered	not applicable . NFPA 13D sprinklers are therefore a voluntary offer , not the exception that relieves a requirement (§5.4)
45	Attached-dwelling visitability (§734)	Applies to attached dwellings only	Detached; 100 % zero-step entries offered voluntarily	not applicable (exceeded)
46	Traffic impact study (Gwinnett DOT <i>Traffic Impact Study Guidelines</i> , 2023)	"A TIS must be submitted concurrently with all zoning applications." The only residential exemption is "no more than seven lots or units." Study level by peak-hour trips: 0–20 → Level 1	41 lots — not exempt ; ≈ 11 –16 PM peak trips → Level 1 (document-only) TIS is required	required — see §6.2. Earlier drafts said "no formal TIS expected"; that is wrong
47	Sewer capacity certification (Gwinnett DWR, form Rev. 07/2023)	Required; PE-sealed; the form carries a "Pre-Rezoning" request type	Request to be filed for 10,670 gpd average daily flow with the Phase 1 gravity route and the Phase 2 question (§5.2)	required — allow 10 business days, plus 20 more if analysis or flow monitoring is needed
48	Accessible parking (2010 ADA Standards §208.2, §502)	1 accessible space where 1–25 spaces are provided; van space 8 ft wide + 8-ft access aisle	1 van-accessible space with an 8-ft access aisle at the clubhouse end of the bay, on the shortest accessible route to the clubhouse entrance (§208.3.1), within the 12 common spaces	appears consistent

4.4 Staff-interpretation questions and the fallback (for the pre-application conference)

- **Q1 — Buffer row of Table 4.1.** Is a detached cottage home on its own fee-simple lot a "detached single-family dwelling in R-2" (0-ft buffer) or one of "all other allowed dwelling types" (20-ft buffer)? The application is designed to the 20-ft row and offers it as a condition either way; the answer affects only how the buffer is recorded.
- **Q2 — §313(1) buffer-within-lot.** May the 20-ft buffer occupy the 20-ft rear yard of each perimeter lot, held in a recorded buffer/conservation easement enforced by the HOA and the City? This is the design basis that lets 100-ft-deep lots sit on both sides of one lane in a 236-ft strip. The section closes exactly — $100 + 35.9 + 100 = 235.9$ ft at the narrowest row — so there is no spare depth, and a survey width below about 229 ft leaves no room for a 22-ft lane with a 2-ft strip and a 5-ft walk. See the caution under item 10.
- **Q3 — Buffer reduction at the entrance.** With a 15-ft southwest curb-return radius, 2.0 ft of the return still lies inside the 20-ft band at the right-of-way corner and the buffer narrows to **18'-0"** over that short length, with 0.1 SF of pavement inside the band. The reduction is requested in the Letter of Intent §5 and stated in voluntary condition 4; is that the form of disclosure staff wants, and does Gwinnett DOT accept a 15-ft return on a minor collector?
- **Q4 — Approved private street.** Is the lane, built to the Site Development Plan Review Checklist's public-street section, an "approved private street" for §319 frontage purposes?
- **Fallback.** If staff requires the buffer in a separate common tract, the lots shorten to 50'-0" $\times$ 82'-0" (4,100 SF, above the 3,000-SF cottage minimum), a concurrent §1005 variance on the 100-ft lot depth is requested under §1005-3.2 (a)–(g), and a

third, shallower house plan is drawn. Lot count, density, buffers and open space are otherwise unchanged.

5. Utilities and infrastructure

5.1 Water

- Source: Gwinnett DWR mains in Arcade Rd (DIP/ACP, 47–107 ft from the front line); nearest hydrant ≈ 61 ft from the NE front corner.
- Proposed: an 8-in DIP main in the lane tract (≈ **1,754 ft**, dead-end) with hydrants at ≈ 400-ft spacing (Gwinnett DWR *Water and Sewer Standards* (April 2016) §2.2.15(b): 350–450 ft, one near the end of the main) — **6 hydrants**. Fire flow for dwellings of this size is 1,000 gpm for 1 hour under IFC 2024 (GA) Appendix B Table B105.1(1), reduced 50 % to **500 gpm** where NFPA 13D sprinklers are installed. DWR may require looping or oversizing the dead-end main; a second tie-in is not physically available, so a flushing hydrant at the terminus and DWR's dead-end criteria apply. Fire-flow test required.
- Individual meters per lot; an irrigation meter for HOA common areas. Average domestic demand **41 × 250 gpd = 10,250 gpd**.

5.2 Sanitary sewer (inverts are GIS attributes; invert survey required)

Phase 1 — gravity to the main already on the property. No extension of any kind.

Item	Value	Basis
Tie-in manhole (existing)	Invert 927.13 ft at (u 272, v –224), on the Arcade Road Townhomes outfall crossing parcel 033	FACTS §2
Phase 1 gravity line	u 980 → u 272 along the lane (708 ft) + a 106-ft lateral to the manhole = 814 ft of 8-in	layout . json path length
Controlling reach	the lane sag (936.3 at u 779) → the manhole, ≈ 613 ft with 5-ft cover at the sag: (936.3 – 5.0 – 927.13) ÷ 613 = 0.68 % available, above the 0.40 % minimum for 8-in	layout . json profile; docs/08 Memo B.3
The hard part	at the lane hump near u 280 (ground 946.4) the pipe is ≈ 17 ft deep . Either accept the deep cut — with rock possible on the ARE unit — or re-route the reach along the SW-edge easement corridor, where it runs at ≈ 10 ft	docs/08 Memo B.3 — to be re-run against the 41-lot alignment
Conclusion, Phase 1	The front 980 ft of lane (24 lots + clubhouse) drains by gravity into the existing 8-in main on the property . Nothing is extended off-site, so the Comprehensive Plan Amendment policy "Do not further extend sewer in this area" (Suburban-Low, p. 73) is not engaged by Phase 1 at all	

Phase 2 — one viable route, and it is not the one earlier drafts called "primary".

Item	Value	Basis
The problem	The lane low point (931.5 ft at u 1,479) lies behind the ridge (953.5 ft at u 1,179); a gravity line back to the Phase 1 system would have to run uphill	layout . json profile
~~In-tract lift station or grinder pumps~~ — foreclosed by policy	Gwinnett DWR <i>Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions</i> (rev. 9/2018): "Private pump stations, private force mains and private gravity sewers are only allowed for commercial properties under single ownership within a development." A 41-lot fee-simple subdivision is not that. A county-owned station is governed by DWR <i>Developer Pump Station Standards</i> (Jan 2014, WSR-24) §1.3.1(A) : stations "will only be permitted when gravity sewer is unavailable ... generally ... more than 5,000 feet down gradient, but this distance can be increased or decreased by GCDWR Division Directors based upon actual field conditions and the size of the project involved" — the Legends at Parkview main is 178 ft away. And there is no Gwinnett standard for grinder pumps or low-pressure sewer at all (zero occurrences in either the 2016 Water and Sewer Standards or WSR-24)	DWR documents
Off-site gravity tie — the only viable route	8-in from the rear low point via (u 1,300, v –114) → (u 1,300, v –235) → the Legends at Parkview manhole at (u ≈ 1,349, v ≈ –406), invert 919.58: ≈ 478 ft total , of which ≈ 178 ft lies off the property , at ≈ 1.45 % — comfortably above the 0.40 % minimum. It needs a recorded sewer easement across one Village Green Ct lot or along the stream corridor	layout . json; docs/08 Memo B
What must be settled	(1) the easement; (2) whether the City reads "Do not further extend sewer in this area" as barring a 178-ft tie to an existing main in the same basin — a policy-interpretation question for the Planning Director and Council; (3) DWR's own view, which should be asked on the "Pre-Rezoning" request type of the Rev. 07/2023 capacity form. Voluntary condition 16 binds this : Phase 2 may be served only by a gravity connection, no private pump station, force main, grinder pump or low-pressure system may be proposed or built, and no land-disturbance permit issues for Phase 2 until the off-site easement is recorded and DWR has certified capacity for the route — failing which the Phase 2 area stays wooded or is re-platted as open space. That is why Phase 1 is designed to stand alone	audit §4.1; data/voluntary-condition 16

Flows — Gwinnett DWR Wastewater Flow Estimation Guidelines (Rev. 10/2022): "Single Family Home 250 per unit" gpd; 175 gpd per 1,000 SF of clubhouse; peaking factor 4.0.

	Units	Average daily flow	Peak (× 4.0)
Whole project	41 DU + 2,400-SF clubhouse	10,670 gpd (41 × 250 + 420 = 10,670; 7.41 gpm)	42,680 gpd = 29.6 gpm

	Units	Average daily flow	Peak (× 4.0)
Phase 1	24 DU + clubhouse	6,420 gpd	25,680 gpd = 17.8 gpm
Phase 2	17 DU	4,250 gpd	17,000 gpd = 11.8 gpm

The "43 × 300 gpd = 12,900 gpd" figure in earlier drafts is retired twice over — the unit count is 41 and 300 gpd per unit appears in no Gwinnett source. The capacity-certification request would be returned on it.

5.3 Stormwater concept (Gwinnett County Stormwater Management Manual via the City; the PE's routing governs)

Method (2025 hydrology sheet HYD-1, Parker Land Design, 04-29-2025): required detention = 10,000 cf per disturbed acre less the required runoff-reduction volume **RRv**; $WQv = 1.2 \cdot (0.05 + 0.009 \cdot I) / 12 \cdot A$, with I = per-cent impervious and A = site area. Earlier drafts of this summary subtracted WQv instead of **RRv**; that is corrected here, and **RRv is carried at zero** at concept stage, as the concept plan states — any runoff-reduction credit the PE earns reduces the requirement.

Quantity	2025 MU filing (3 basins, HYD-1)	Cottage plan (this application)
Basin / site area	424,548 SF (9.75 ac)	417,174 SF (9.577 ac GIS)
Impervious area	235,388 SF = 5.41 ac (55.4 %)	184,148 SF = 4.23 ac (44.1 %)
$Rv = 0.05 + 0.009 \cdot I$	0.50–0.56	0.447
$WQr = 1.2 \cdot Rv$ (in)	0.60–0.67	0.537
WQv	23,293 cf	18,659 cf
RRv (1.0-in basis), available but not taken	—	15,552 cf
Disturbed area	whole basins	328,484 SF = 7.541 ac
Required detention (10,000 cf/ac, RRv = 0)	74,167 cf	75,410 cf
Provided detention	(3 basins)	77,863 cf — 103.3 % of the requirement

- **The disturbed-area correction is the single biggest arithmetic change in this package, and it is not in the applicant's favour.** The concept plan previously assumed ≈ **4.4 disturbed acres**; the real figure, rastered at 2 ft, is **7.541 ac**. Site raster area 418,228 SF less the preserved union of the 20-ft perimeter buffer bands (73,744 SF), the net creek-woods tract (16,000 SF) and the 50-ft undisturbed stream buffer (0 SF additional — it lies wholly inside the other two) = 89,744 SF preserved, leaving **328,484 SF disturbed**. The raster site area runs 0.25 % above the 417,174-SF shoelace because of 2-ft quantisation, so the disturbed area is if anything conservative. Required detention rose with it, from ≈ 44,000 cf to **75,410 cf — more than the 2025 mixed-use filing's 74,167 cf**, even though this plan is 21.8 % less impervious, because the 2025 sheet took its volume on three smaller drainage basins rather than on the whole disturbed site.
- **The basins were enlarged to meet it, and that cost a lot.** Pond 1 (SW side, basin u 790–970, top of bank **52 ft × 180 ft**) holds **33,129 cf**; Pond 2 (SW side, basin u 1,490–1,670, **64 ft × 180 ft**) holds **44,734 cf — 77,863 cf** total, each basin 6 ft deep with 3:1 side slopes, prismatic volume between top of bank and basin bottom. Pond 2's tract had to grow from 15,000 to 20,000 SF, which is where the second of the two lost lots went. An independent prismatic check in `tools/dev_summary_calcs.py` returns 33,696 and 45,360 cf for the same rectangles — about 1.5 % above the generator's figures, which is the margin of the end-slope treatment. The PE's routing governs either way.
- **Where the basins sit relative to the buffers.** Pond 1's top of bank is **41.5 ft** from the SW property line and Pond 2's is **30.0 ft**, so neither now encroaches the 20-ft perimeter buffer (the earlier layout put Pond 2's bank 0.1 ft inside it). The nearest pond top of bank is **80.2 ft** from a digitized stream centerline, i.e. 75.2 ft after the 5.0-ft top-of-bank allowance. **Rock probes come first:** both ponds sit on the ARE map unit (gneiss at 22–40 in), and a 6-ft basin is exactly where that matters.
- Runoff-reduction practices (bioretention with underdrains on clay subsoils, pervious guest parking) are favoured because the receiving Jackson Creek headwaters are 303(d)-listed for urban runoff; every cubic foot of **RRv** the PE can document comes straight off the 75,410-cf requirement (at the 1.0-in basis, 15,552 cf of it).
- The 2025 rule is used here only as a screening yardstick. **The PE's GCSWMM routing (WQv , CPv , $Qp25$, $Qf100$) governs**, and every disturbed acre carries the LDP fee (\$40/disturbed ac) and the NPDES fee (\$80/ac) — on 7.541 ac, not on 4.4.
- **Water quality:** $WQv = 1.2 \times 0.447 \div 12 \times 417,174 \text{ SF} = 18,659 \text{ cf}$, treated in the two basins. Recomputed independently from the lot polygons the same formula gives **18,662 cf**; the 3-cf difference is the 30-SF difference in the impervious take-off noted in §3.7.

5.4 Fire protection and emergency access (Gwinnett County Fire; IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04)

- **The one live issue is the dead end.** The lane is a single **1,754-ft** dead end. Appendix **D103.4** Table: 0–150 ft / 20 ft wide / no turnaround; 151–500 / 20 ft; 501–750 / **26 ft**; **over 750 ft** → **"Special approval required."** The applicant requests that special approval and discloses that the lane is 22'-0" wide against the 26 ft the table asks in the band below.
- **The turnaround intervals now have room.** The three hammerheads fall at stations 594.5, 1,164.5 and 1,744.5 ft, i.e. intervals of **594.5 / 570.0 / 580.0 ft** — the longest is 155 ft inside the 750-ft trigger. The earlier layout's first interval was 746.4 ft, 3.6 ft inside it, which no survey shift could be allowed to disturb; that fragility is gone.
- **The two-access requirement is not triggered.** The model IFC 2024 Appendix D107.1 sets the threshold at 30 dwelling units; **Georgia replaces it with 120** — Ga. Comp. R. & Regs. 120-3-3-.04, Modifications to Chapter 5, item 7: "Developments of one- or two-family dwellings where the number of dwelling units **exceeds 120** shall be provided with two separate and approved fire apparatus access roads." At **41 units** D107.1 does not apply, and a second access is in any case physically impossible — no street abuts the rear or either long side.
- **NFPA 13D sprinklers in every home are therefore a voluntary offer, not a code requirement** (voluntary condition 5). They are worth offering: they halve the required fire flow on a 1,754-ft dead-end main (1,000 → 500 gpm, IFC 2024 (GA) Appendix B105.1 exception), and they are the natural consideration to put on the table for the D103.4 special approval and the 22-ft width. At roughly \$3,000 per home they are a **≈ \$123,000** election at 41 units, and the Fire Marshal — not this document — should decide whether they buy the width relief.
- **Other mitigation offered:** a 22-ft clear lane with no on-lane parking (signed and enforced by the HOA); three 20 ft × 60 ft hammerhead turnarounds; hydrants in the lane; a sprinklered clubhouse as the building code requires (NFPA 13 or 13R — VERIFY).
- Georgia's D107.1 Exception 3 also supplies a named menu for developments with fewer than two entrances — enhanced turning radii, increased road widths, D103.6 fire-lane signage, an additional non-vehicular all-weather access designed to Appendix D loading, or a written statement by the Fire Code Official — and O.C.G.A. §25-2-12(e)(4) provides a waiver route on the Safety Fire Commissioner's form, on which "the State Fire Marshal or designated representative shall respond within 30 days" and plans are **deemed approved** if that clock is missed.
- **First-due:** Station 3, **≈ 2.9 mi / ≈ 6 min** free-flow (brief §4.6). Gwinnett Fire Marshal review is requested at the pre-application stage.

5.5 Refuse, electric, gas, communications

- Refuse: individual roll-carts collected curbside on the lane by the city's hauler; the 22-ft lane and T-turnarounds accommodate a rear-loader (turning template by the PE — VERIFY the hauler's vehicle). No dumpster pad.
- Electric, gas, telecommunications: underground in the lane tract (Jackson EMC / Georgia Power service territory VERIFY; Atlanta Gas Light); generator-ready clubhouse as a warming/cooling centre.
- Street lighting: full-cutoff, **≤ 3000 K**, house-side shields on fixtures facing R-1 lots (voluntary condition 11).

6. Traffic

6.1 Trip generation (ITE *Trip Generation Manual*, 11th ed.)

The 11th edition renamed and re-scoped **LUC 251** to "Senior Adult Housing – Single-Family" (merging the former detached and attached data sets), and the manual itself is paywalled. Four secondary compilations were reconciled on 2026-09-03 and none is demonstrably an 11th-edition source; the honest presentation is a **rate band**, with the note that the traffic engineer must pull the manual page before filing.

Scenario	Units	LUC	Daily	AM peak	PM peak
Proposed — 55+ detached cottages	41 DU	251, rate band 3.70–4.31 daily / 0.20–0.34 AM / 0.26–0.39 PM per DU	152–177	8.2–13.9	10.7–16.0
Same, at the FACTS \$3 rate (0.30 / 0.30)	41 DU	251	≈ 177	12.3	12.3
The same 41 lots, unrestricted single-family	41 DU	210 Single-Family Detached (9.43 / 0.70 / 0.94)	≈ 387	28.7	38.5
R-1 as currently zoned, theoretical maximum (4 du/ac × 9.44 ac = 37 lots; the real R-1 yield on a 236-ft strip would be lower)	37 DU	210	≈ 349	25.9	34.8
RZ-2025-01 MU filing (no TIS was filed; order-of-magnitude, before pass-by or internal capture)	104 condos + 24,400 SF retail/service/office	220 Multifamily Low-Rise + 822 Strip Retail Plaza	≈ 2,030	—	≈ 214

- **The conclusion does not turn on which rate is used.** At the highest rate found (0.34 AM / 0.39 PM) and at the top of the offered unit range (47 units — the cap in voluntary condition 2, six above the 41 now drawn), the PM peak is **18.3 trips** — still inside Gwinnett DOT's 0–20 band.
- The cottage program generates roughly **one-third** of the PM peak of the same lots without an age restriction (12.3 against 38.5) and about **5.8 %** of the 2025 mixed-use concept.

6.2 Findings

- **A Level 1 TIS is required, and this application will file one.** Gwinnett DOT *Traffic Impact Study Guidelines* (2023): "A TIS must be submitted concurrently with all zoning applications and land disturbance development," and the only residential exemption is "single-family detached and townhome residential development consisting of **no more than seven lots or units.**" At 41 lots the project is not exempt; at 0–20 peak-hour trips the study level is **Level 1 (document-only)** — location description, existing and proposed land use, trip generation, access-management review, adjacent access spacing upstream and downstream, intersection sight distance, connectivity and circulation, functional classification, posted speed and future identified projects. Operations analysis, turn-lane warrant analysis and signal warrants begin at Level 2 and are **not** required. Note that these guidelines are written for unincorporated Gwinnett County under UDO §270-20.8; the City's own requirement is to be confirmed at the pre-application conference, and Gwinnett DOT will apply them at the driveway-permit stage on its own road.
- **New traffic counts are required.** The 14,800 AADT figure comes from GDOT station 135-6689 in **2017**. The same Gwinnett DOT guidelines require counts "less than two years old" from GCDOT or GDOT and, failing that, that "the developer shall be responsible for obtaining new, 24-hour counts, from an approved contractor, as directed by GCDOT." A 24-hour count is budgeted.
- **Left-turn lane not warranted.** Gwinnett DOT *Criteria and Guidelines for Left Turn Lanes* (June 20, 2007), **Table 1**, residential developments, 2-lane route, posted 30–35 mph, ADT ≥ 6,000: the threshold is **75 lots**. At 41 lots no left-turn lane is warranted (to be confirmed with the new count).
- **Spacing and sight distance.** Gwinnett UDO §900-40.5, **Table 900.3**: minimum driveway separation at 35 mph is **244 ft**, centerline to centerline; provided **251.2 ft** from the Arcadia Place centerline (250.7 ft chord), a margin of 7.2 ft. Gwinnett UDO §900-40.4, **Table 900.2**: intersection sight distance at 35 mph on a 2-lane road is **390 ft** each way, on a curved frontage — **field survey required**. §900-40.5(C) limits the project to one driveway per 400 ft of frontage, i.e. one.
- **Pedestrian and transit.** 5-ft sidewalk along the Arcado Rd frontage; Ride Gwinnett Route 25 stops at Killian Hill Rd @ Arcado Rd, 0.30 mi away (no continuous sidewalk verified — the applicant requests county and city gap closure); the ADA paratransit band covers the site.
- **School buses:** none (HOPA 55+).

7. Schools (Gwinnett County Public Schools — Camp Creek ES / Trickum MS / Parkview HS cluster)

Measure	RZ-2025-01 (104 condos; GCPS planning estimate, July 2025, 4817_school-impact-analysis.pdf)	This application (41 HOPA 55+ cottages)
Camp Creek ES (capacity 1,325; forecast 1,414 → 1,383, over capacity by 89 → 58)	+25 students	0
Trickum MS (capacity 2,275; forecast 2,203 → 2,104, under capacity)	+13	0
Parkview HS (capacity 2,900; forecast 3,148 → 3,093, over capacity by 248 → 193)	+19	0
Total	+57	0

Housing for older persons under 42 U.S.C. §3607(b)(2)(C) — at least 80 % of occupied units with a resident 55 or older, which at 41 units means **33 units** ($0.80 \times 41 = 32.8$, rounded into the 80 percent by 24 CFR §100.305(g)) — generates no school-age children. The two over-capacity schools in the cluster receive no added enrolment, and the §1003-7(4) "burdensome use of ... schools" criterion is answered in the negative. VERIFY the covenant's minor-occupancy clause against HUD guidance; note also 24 CFR §100.306(b), which makes phrases such as "adult living" or "adult community" inconsistent with 55+ intent — the marketing language is bound in the covenants.

8. Fiscal and community effects

8.1 Tax base (illustrative; market figures VERIFY with a broker CMA and the Gwinnett Tax Commissioner once 2026 rates are adopted)

Item	Low	High	Basis
Sale price per cottage (assumed)	\$420,000	\$460,000	positioning between Cottages at Noble Village resales (\$342,500–\$414,900) and Snellville detached 55+ product (≈ \$447k Saddlebrook; mid-\$500s Soleil) — VERIFY
Completed value, 41 homes	\$17.22 M	\$18.86 M	41 × price
Existing 2026 FMV of the four parcels	\$1,267,600		2026 digest
Assessed value (40 %)	\$6.89 M	\$7.54 M	O.C.G.A. 40 % ratio
City of Lilburn tax at 5.43 mills (2026 rate set by Resolution No. 2026-11, calendared 2026-08-31)	≈ \$37,400 /yr	≈ \$41,000 /yr	before homestead exemptions; existing ≈ \$2,750/yr
City tax if every owner claimed S1R + L5 + the 2026 \$40,000 senior exemption (\$60,000 assessed each)	≈ \$24,000 /yr	—	illustrative floor: (6.89 M – 41 × \$60,000) × 5.43 ÷ 1,000
County + school levies at 31.6 mills (2025 rates: county 6.95 + fire 3.20 + recreation 1.00 + GCPS 18.70 + bond 1.45)	≈ \$218,000 /yr	≈ \$238,000 /yr	before exemptions; the school share (20.15 mills, ≈ \$139k–\$152k) is removed for owners 65+ under the LSA income cap (\$124,648 in 2026)

The community adds roughly **fourteen to fifteen times** the current taxable value of the assemblage (13.6× at the low price, 14.9× at the high) while sending **no children** to GCPS; the senior exemptions that trim the school levy are the same exemptions the neighbours already enjoy. Two fewer lots take about **\$840,000–\$920,000** off the completed value against the 43-lot draft — worth stating, because it is the fiscal cost of the stream screen and the basin enlargement.

8.2 HOA and maintenance

- Mandatory homeowners' association (Georgia Property Owners' Association Act, O.C.G.A. §44-3-220 et seq.) owning and maintaining the lane tract, sidewalks, ponds, buffers, creek woods, amenity tract, entry sign and lighting, and providing yard maintenance for every lot.
- The line-item budget is now built at 41 lots** (tools/hoa_budget.py → data/hoa-budget-derived.json, carried in docs/13): operating \$123,356 + reserves \$40,451 + contingency \$6,168 + bad-debt allowance \$3,399 = **\$173,374 a year, \$4,229 per lot per year = \$352 per lot per month** (operating \$251 + reserves \$82 + contingency and bad debt \$19), with curbside refuse a further \$22 if the association contracts it. **During Phase 1 alone, 24 lots carry nearly the same fixed plant, so the assessment is \$449 per lot per month** — a number the declarant must either subsidise or disclose to the first buyers.
- The "\$200–\$350/month" range quoted in earlier drafts is retired: it was not supported by a budget, and the real figure sits at the top of it at full build-out and above it during Phase 1. Comparables: Cottages at Noble Village \$93–\$280/mo; Soleil Summit Chase \$389/mo.

8.3 Demand references (market data — VERIFY; commercial sources cited in

docs/11-site-context-brief.md §5)

- Census tract 504.61 is 35.5 % aged 55+, 90 % owner-occupied, 67 % pre-1980 housing (ACS 2020–2024); ZCTA 30047 has ≈ 8,500 owner-householders 55+ (25.8 % of 70,121 residents are 55+; median household income \$88,652).
- Lilburn's only 55+ product, Cottages at Noble Village (1,225–1,450 SF), is sold out; no detached 55+ community exists inside the city. The nearest detached competitors are in Snellville.
- ARC Series 17 (2024) forecasts Gwinnett growing to 1.20 M by 2050 with residents 75+ rising to ≈ 12 % of the region.
- Lilburn ZHVI \$401,137 (June 2026, –2.6 % YoY); King David Manor median FMV \$349,200; Legends at Parkview \$693,500 — the cottages price between the two abutting subdivisions.

8.4 Comprehensive-plan fit

Single-family detached housing at **4.34 du/ac** inside the adopted "Established Residential" character area (2024 plan, printed p. 35) and, under the pending 2026 Amendment, the "Suburban-Low" class, where Single-Family Cottages and Single-Family are *Appropriate* and the 2025 mixed-use concept is *Not Appropriate*; below R-2's 8 du/ac maximum and far below the 2025 filing's ≈ 11 du/ac. The Amendment's stated policies are addressed directly: one-story massing answers "encourage new developments to match existing design character"; frontage and lane sidewalks answer "require sidewalks"; and **Phase 1 extends no sewer at all**, using the main already on the property, while Phase 2's route is left open for the City and DWR to settle (§5.2).

9. Voluntary zoning conditions offered (to be restated in the Letter of Intent and recorded in the declaration of covenants)

1. **Age restriction.** The development shall be operated as housing for older persons (55+) under 42 U.S.C. §3607(b)(2)(C) and 24 CFR §100.300–.308, with a recorded declaration, published policies and biennial verification (24 CFR §100.307(c)), enforceable by the City of Lilburn. At the 41 units shown, the 80-percent occupancy test of 24 CFR §100.305(a) together with the fractional-unit rule of §100.305(g) ($0.80 \times 41 = 32.8$, and the fraction counts into the 80 percent) requires **at least 33 of the 41 units** to be occupied by at least one person aged 55 or older, leaving at most 8 units that need not be; the declaration shall state that figure and shall recompute it if the unit count changes. Advertising and community documents shall not use "adult living," "adult community" or similar phrases (24 CFR §100.306(b)).
2. **Maximum units.** Not more than **47** detached single-family cottage homes (**41 shown**); no attached, multifamily, commercial or non-residential use other than the clubhouse and amenities serving residents.
3. **Height.** All dwellings one story; maximum ridge height 24 ft above finished grade (the plans as drawn on Sheets A-2.1 and A-2.2 are 17'-10" for Plan A and 19'-0" for Plan B above finished floor, i.e. 18'-6" and 19'-8" above finished grade at the front, finished floor being 8 in above grade); clubhouse one story.
4. **Buffers.** A 20-ft undisturbed buffer along all property lines abutting R-1 property (SW, NE and NW lines), held in a recorded buffer/conservation easement where it lies within lots and in common tracts elsewhere; supplemental evergreen planting where existing vegetation is sparse; no structures, fences or grading in the buffer except (a) approved pond and utility encroachments not exceeding 50 ft in perpendicular width and (b) **the single buffer reduction requested in the Letter of Intent §5 and dimensioned on Sheet C-2.3 (entry enlargement, to be issued with the filed set) — at the tangent point of the southwest curb return of the entrance on the Arcade Road right-of-way, where the buffer narrows to a minimum of 18'-0" instead of 20'-0", a reduction of 2.0 ft measured along the right-of-way affecting 0.1 SF of pavement.** The buffer is at its full 20-ft width everywhere else on the southwest, northeast and northwest lines. No other buffer reduction is requested, and none shall be taken without an amendment to these conditions.
5. **Fire sprinklers — voluntary offer.** Every dwelling shall be equipped with an NFPA 13D residential sprinkler system and the clubhouse sprinklered as the building code requires, with systems maintained under the HOA covenants. This is offered **voluntarily**: at 41 units the two-access requirement of IFC 2024 Appendix D107.1 as modified by Ga. Comp. R. & Regs. 120-3-3-.04 (which raises the trigger to more than 120 dwelling units) is not engaged, and sprinklers are therefore additional protection and a 50-percent fire-flow reduction (IFC 2024 Appendix B §B105.1), not a required mitigation. The applicant asks that this condition be accepted as offered; if the Gwinnett County Fire Marshal states in writing that sprinklers are unnecessary, the condition may be released by the Fire Marshal's written concurrence and City staff's approval without a further public hearing.
6. **Single access.** One driveway on Arcade Rd, located in the SW third of the frontage; no gate.
7. **No Arcade Road driveways.** No individual lot shall have direct vehicular access to Arcade Rd.
8. **Architectural standards.** Exterior walls of fiber-cement lap and board-and-batten siding with brick or stone water table/piers; architectural-shingle roofs with 8:12 main and front-wing gables, the front-wing gable carried forward to roof the covered front porch, and a 6:12 gable over the garage; front-loaded garages recessed ≥ 5 ft behind the front wall; covered front porches ≥ 6 ft deep; zero-step entries; floor plans, elevations and materials substantially as shown on the sheets submitted with the application — Sheets A-1.1 and A-1.2 (floor plans, Plans A and B), A-2.1 and A-2.2 (exterior elevations, Plans A and B), A-2.3 (exterior colour schemes and materials), A-3.0 (clubhouse, mail kiosk and entry monument sign) and A-4.0 and A-4.1 (typical lots); three approved colour schemes; no wall signs on any dwelling.
9. **Stream buffer.** The stream head and its 50-ft undisturbed buffer and 75-ft impervious setback preserved in a common "creek woods" tract, and the southwest area at u 1,230–1,280 that does not clear the stream screen preserved unlotted as a common stream-setback tract with no impervious cover; no lot, structure or pavement within 75 ft of the top of bank as field-delineated. The field study shall establish first whether and where a regulated stream begins (spring/seep or the 25-acre drainage-area point) and then the top of bank, and the plat shall show the 25-ft state buffer (O.C.G.A. §12-7-6(b)(15)) and the 50-ft/75-ft city lines separately.
10. **HOA maintenance.** A mandatory HOA owning and maintaining the private lane, sidewalks, stormwater facilities, buffers, common tracts, entry sign and lighting, with yard maintenance for all lots.
11. **Lighting.** Full-cutoff fixtures ≤ 3000 K; lane lighting mounted ≤ 14 ft with house-side shields facing R-1 lots; no court lighting for the pickleball courts.
12. **Construction hours.** Exterior construction limited to 7:00 AM – 7:00 PM Monday–Friday and 8:00 AM – 5:00 PM Saturday; none on Sundays or federal holidays (city ordinance governs if stricter).
13. **Tree survey and preservation.** A tree survey and preservation plan meeting Code Ch. 109 Art. V (16 TDU/ac) submitted with the LDP; preserved trees ≥ 12 -in DBH in the buffers, creek woods and stream-setback tract credited; phased clearing with immediate stabilization.

14. **Entry sign.** One monument sign ≤ 32 sf, ≤ 6 ft high, externally lit, standing about 19 ft from the Arcade Road right-of-way within the front amenity tract and outside the sight triangle; permitted separately under the sign regulations.
15. **Phasing.** Development in two phases at the **u $\approx$ 980-ft line** as shown on Sheet C-6.0 (phasing, block and open space exhibit, to be issued with the filed set); Phase 1 (24 lots) shall include the amenity block, Pond 1 and the first hammerhead turnaround shown on Sheet C-2.0. Phase 2 (17 lots) shall not commence until condition 16 is satisfied, and the existing dwelling at 4535 Arcade Road, which stands on the proposed lane alignment about 1,200 ft back from the right-of-way, shall be removed under permit before any Phase 2 land disturbance.
16. **Sanitary sewer.** Phase 1 shall be served by gravity to the existing 8-in Gwinnett County main already on the property (the Arcade Road Townhomes outfall, manhole invert 927.13), and requires **no sewer extension**. Phase 2 shall be served **only by a gravity connection**; no private pump station, private force main, private gravity sewer, grinder pump or low-pressure sewer system shall be proposed or built (Gwinnett DWR *Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions*, rev. 9/2018, allows private facilities "only ... for commercial properties under single ownership within a development"). No land-disturbance permit shall issue for Phase 2 until (a) a permanent off-site sanitary sewer easement for the gravity connection is recorded and (b) Gwinnett DWR has issued a sewer capacity certification covering the Phase-2 route. If either is not obtained, Phase 2 shall not be built and the Phase-2 area shall remain in its existing wooded condition or be re-platted as open space.
17. **Traffic impact study and access management.** The applicant shall submit a **Level-1 traffic impact study** prepared to Gwinnett DOT *Traffic Impact Study Guidelines* (2023) with the zoning application, based on **new 24-hour traffic counts** on Arcade Rd obtained from an approved contractor (the 2017 GDOT count does not satisfy the guidelines' two-year currency rule); shall furnish an intersection-sight-distance certification by a Georgia Professional Engineer for the single entrance (390 ft each way at 35 mph, Gwinnett UDO §900-40.4, Table 900.2); and shall obtain a Gwinnett County DOT driveway/encroachment permit before any land disturbance.
18. **Sidewalks and transit.** A 5-ft sidewalk along the full Arcade Rd frontage and 5-ft sidewalks within the lane tract as shown — on the northeast side for the full length of the lane and on the southwest side from u 530, where the tract first reaches the 36 ft that two walks require; the applicant will cooperate with the county and city on closing the sidewalk gap to the Killian Hill Rd bus stop.

10. Open items / VERIFY list

#	Item	What settles it
1	Boundary, acreage (9.44 deeded vs 9.577 GIS), metes-and-bounds legal description	Georgia RLS boundary survey and sealed legal description
2	Site width — there is no longer any margin in the section. 100-ft lots on both sides plus a 35.9-ft lane tract exactly consume the 235.9-ft width at the narrowest lot row, and the tract is the residual: every foot the survey removes comes off the lane. At 229 ft of width the tract is down to the bare 29.0 ft that pavement, one strip and one walk require	RLS survey (a 230-ft width forces the 50'-0" $\times$ 82'-0" fallback and a third house plan)
3	Q1 and Q2 buffer interpretations (Table 4.1 buffer row; §313(1) buffer-within-lot)	Pre-application conference with the Planning Director (Reid Turner, 770-279-3715). The Development Regulations §5.9 open-space question is closed — §5.9 is "Recreation Areas" and applies only at 50 acres or more
4	Plan B's ridge: the data and the drawing differ by 4 in. Issued Sheet A-2.2 draws 19'-8" above finished grade (19'-0" above the floor); data/plans.json computes 19'-4" (19.33 ft) above the floor from a 29.5-ft main-gable span, i.e. 20'-0" above grade. This summary and condition 3 state the drawn figures, because a recorded condition must not cite a dimension the filed sheet does not draw	Architect reconciles the roof model; conform data/plans.json (docs/09 §5.3 item G-1) or re-issue A-2.2, before filing . Plan A is consistent at 17'-10" / 18'-6"
5	Buffer reduction at the SW curb return — now consistent everywhere. The drawing, the Letter of Intent §5 and voluntary condition 4 all state the same figure: the buffer narrows to 18'-0" instead of 20'-0" at the tangent point, a 2.0-ft reduction affecting 0.1 SF of pavement. What remains is Gwinnett DOT's concurrence on the 15-ft return radius and the entry enlargement sheet	Gwinnett DOT driveway permit; Sheet C-2.3 to be issued with the filed set
6	Stream: whether the reach is a regulated "stream" at all (the 25-acre drainage-area test) and where its top of bank lies. The screen against all three reaches now holds every element at ≥ 80 ft from a digitized centerline (75.2 ft after the 5-ft allowance), and it cost the u 1,230–1,280 lot	Field study by a qualified professional (GA EPD <i>Field Guide</i> , Sept 2017, for the state-waters half) + City review authority. A favourable delineation could return that lot; an unfavourable one could take more
7	Sewer inverts (927.13 at the on-site MH; 919.58 at the Legends MH); the Phase 2 easement across a Village Green Ct lot; the City's reading of "Do not further extend sewer in this area"	Invert survey; Gwinnett DWR sewer capacity certification on the "Pre-Rezoning" request type at 10,670 gpd; Planning Director and Council on the policy question

#	Item	What settles it
8	Comprehensive Plan Amendment adoption date (available 9/14, 10/12 or 11/9/2026) and which Future Development class the four parcels fall in (Suburban-Low or Suburban-Medium — the sewer policies are opposite)	City Council adoption record; Planning Director in writing
9	Stormwater is now sized, but on a screening rule and on unprobed ground. 77,863 cf provided against 75,410 cf required on 7.541 disturbed acres — a 3.3 % margin, with RRv carried at zero. Both 6-ft basins sit on the ARE map unit (gneiss at 22–40 in)	PE hydrology per GCSWMM; rock probes / geotechnical report before the volumes are fixed ; any RRv credit reduces the requirement
10	Soil map-unit names and ratings (§2.4)	NRCS Web Soil Survey AOI report (Soil Data Access not reachable 2026-08-29)
11	ITE 11th-ed. LUC 251 rates (the band 0.20–0.34 AM / 0.26–0.39 PM); the Level 1 TIS itself; a new 24-hour count on Arcade Rd	The traffic engineer's copy of the manual; a count from a GCDOT-approved contractor; Gwinnett DOT pre-application
12	IFC 2024 (GA) Appendix D103.4 special approval for the 1,754-ft dead end at 22-ft width; hydrant spacing; refuse-vehicle turning	Gwinnett County Fire Marshal (Ga. Comp. R. & Regs. 120-3-3-.04; O.C.G.A. §25-2-12(e)(4) waiver route); city hauler
13	Water: dead-end main looping/oversizing, fire-flow test	Gwinnett DWR
14	Tax figures — 2026 city millage, GCPS 2026 rate and sale prices (\$420–460k). The HOA budget is no longer an open item: data/hoa-budget-derived.json carries a line-item 41-lot budget at \$352/lot/month full build-out and \$449 during Phase 1	Tax Commissioner; broker CMA; the declarant's decision on subsidising Phase-1 assessments
15	Formal disposition of RZ-2025-01 (tabled/dormant vs withdrawn)	Written confirmation from the Planning Director
16	Demolition permits: 4541 Arcade Rd in Phase 1 and 4535 Arcade Rd, which stands on the lane alignment at u ≈ 1,202 , in Phase 2; 4535's septic-vs-sewer status	Owner records; Gwinnett DWR; city permits
17	HOPA covenant terms (minor-occupancy clause, verification procedure, the 24 CFR §100.306(b) marketing-language bar); the declaration must state the 33-of-41 figure and recompute it if the count changes	Counsel; HUD 24 CFR Part 100 Subpart E
18	Sidewalk on the SW side for u < 530 (6 of the 14 SW lots) — a walk on both sides needs a 36.0-ft tract and the narrowest is 35.9 ft	Pre-application answer, recorded; 0.5 ft of lot depth or a 1.5-ft strip would close it
19	Tree-canopy inventory; USFWS IPaC screen	Arborist / landscape architect at LDP
20	The offered unit cap now sits 6 units above the drawing. Voluntary condition 2 caps the project at 47 units, a figure set when the plan showed 43; it shows 41. A cap that far above the plan invites the question of what a 47-lot version would look like, and the stream screen and basin sizes suggest it no longer exists	Owner's decision before filing — lower the cap to 43 or 44, or be ready to explain the difference at the hearing
21	Documents outside this summary that still carry 43-lot arithmetic must be re-run behind the regenerated layout — docs/03 §§2–6, docs/08 Memos A–E, docs/10 §4 and docs/13's budget tables. tools/dev_summary_calcs.py and tools/hoa_budget.py have been re-run; tools/memo_calcs.py has not been checked by this pass	Re-run each generator, then python3 verify/check_dev_summary.py, python3 tools/check_docs.py and python3 verify/check_sheet_refs.py

Closed since the previous draft: the block-length question (594 / 570 / 580 ft, all inside Table 4.2's 600 ft, so no waiver is sought); the 746-ft first turnaround interval (now 594.5 ft, with room against the 750-ft D103.4 trigger); Pond 2's 0.1-ft encroachment into the perimeter buffer (the basins were re-shaped and now stand 30.0 and 41.5 ft off the SW line); the "lot 10" stream conflict (that ground is now an unlotted stream-setback tract); the open-space residual (the enumerated tracts now close exactly on 141,089 SF with no unallocated slivers); the "two older houses near the front" description of the existing improvements (only 4541 is near the front — 4535 stands on the lane 1,200 ft back); the document-versus-drawing contradiction on the entrance buffer reduction (all three now say 18'-0" / 2.0 ft / 0.1 SF); the "≈ 4.4 disturbed acres" assumption (7.541 ac); the Development Regulations §5.9 open-space question; the "garage inside the body" floor-area alternative; the fire-code edition (IFC 2024 as modified by Ga. Comp. R. & Regs. 120-3-3-.04, never the 2018 edition); and the two-access requirement, which Georgia does not trigger at 41 units.

Prepared by: owner/applicant (self-prepared DRAFT per the Lilburn 2026 Application Instructions); to be reviewed and, where required, sealed by a Georgia RLS (boundary and legal description), PE (sewer, stormwater, lane, traffic), architect (plans and elevations) and landscape architect (tree and buffer plans) before filing.

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.